



SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

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ASSESSMENT TOOL -1

CO2_AT1

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1. A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three different labs using fixed subnet masks. Lab A is assigned a /26 subnet, Lab B is assigned a /27 subnet, and Lab C is assigned a /28 subnet. Based on the given subnet masks, apply subnetting concepts to determine the network address of each lab. Further, identify the valid range of host IP addresses and the broadcast address for each subnet. Finally, calculate the total number of usable host IP addresses available in each lab's subnet.

ANSWER :-

Variable Length Subnetting (VLSM)

When subnets with different prefix lengths (/26, /27, /28) are allocated from a single base block (192.168.20.0/24), subnets are allocated sequentially starting from the largest block to avoid address overlapping.

Calculations & Results

1. Lab A (/26 Subnet)

- Subnet Mask: 255.255.255.192
- Block Size: $2^{32-26} = 64$
- Network Address: 192.168.20.0
- Valid Host IP Range: 192.168.20.1 to 192.168.20.62
- Broadcast Address: $192.168.20.(0 + 64 - 1) = 192.168.20.63$
- Usable Hosts: $2^{32-26} - 2 = 64 - 2 = 62$ hosts

2. Lab B (/27 Subnet)

- Subnet Mask: 255.255.255.224
- Block Size: $2^{32-27} = 32$
- Network Address: 192.168.20.64 (starts immediately after Lab A's broadcast)
- Valid Host IP Range: 192.168.20.65 to 192.168.20.94
- Broadcast Address: $192.168.20.(64 + 32 - 1) = 192.168.20.95$

- Usable Hosts: $2^{32-27} - 2 = 32 - 2 = 30$ hosts

3. Lab C (/28 Subnet)

- Subnet Mask: 255.255.255.240
- Block Size: $2^{32-28} = 16$
- Network Address: 192.168.20.96 (starts immediately after Lab B's broadcast)
- Valid Host IP Range: 192.168.20.97 to 192.168.20.110
- Broadcast Address: 192.168.20. (96 + 16 - 1) = 192.168.20.111
- Usable Hosts: $2^{32-28} - 2 = 16 - 2 = 14$ hosts

2. A network administrator has configured two systems with the following details: System 1 has IP address 192.168.1.130 with subnet mask 255.255.255.192, and System 2 has IP address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system. Identify the valid host IP range for both subnets based on the given subnet mask. Calculate whether each IP address falls within its respective subnet range. Also, determine the number of usable host addresses available in each subnet.

ANSWER :-

Host Subnet Verification

To identify which subnet an IP address belongs to using a subnet mask of 255.255.255.192 (/26), perform a bitwise AND operation between the host IP address and the subnet mask.

- Subnet Mask 255.255.255.192 gives a Block Size of $256 - 192 = 64$.
- Subnet boundaries occur at multiples of 64: 0, 64, 128, 192.

Calculations & Results

1. System 1 (192.168.1.130)

- Subnet Boundary: Since $128 \leq 130 \leq 191$, it belongs to the 192.168.1.128/26 block.
- Network Address: 192.168.1.128

- Broadcast Address: 192.168.1.191
- Valid Host Range: 192.168.1.129 to 192.168.1.190
- Placement Verification: 192.168.1.130 lies inside 192.168.1.129–190. It is a valid assignable host IP.
- Usable Hosts: $64 - 2 = 62$ hosts

2. System 2 (192.168.1.190)

- Subnet Boundary: Since $128 \leq 190 \leq 191$, it belongs to the same 192.168.1.128/26 block.
- Network Address: 192.168.1.128
- Broadcast Address: 192.168.1.191
- Valid Host Range: 192.168.1.129 to 192.168.1.190
- Placement Verification: 192.168.1.190 lies inside 192.168.1.129–190. It is a valid host IP (the last usable address in this subnet).
- Usable Hosts: $64 - 2 = 62$ hosts

3. An organization has been allocated the IP block 172.16.0.0/16 for its internal network. The network administrator has already decided the subnet masks for each department as follows: HR → /25, Sales → /26, IT → /27, and Admin → /28. Using the given subnet masks, apply subnetting techniques to assign suitable subnet addresses for each department. Determine the network address, valid host IP range, and broadcast address for each subnet. Based on the allocation, calculate the number of usable host addresses in each department. Finally, identify the remaining unused IP address range within the given network block.

ANSWER :-

Department Address Allocation & Unused Range Assigning subnets of varying sizes to departments sequentially minimizes wasted IP address space. The base block 172.16.0.0/16 provides 65,536 total addresses. Allocating subnets starting from 172.16.0.0 leaves the remaining space contiguous.

Subnet Allocations

- HR Department (/25)
 - Network Address: 172.16.0.0
 - Valid Host Range: 172.16.0.1 – 172.16.0.126
 - Broadcast Address: 172.16.0.127
 - Usable Hosts: $2^7 - 2 = 126$ hosts
- Sales Department (/26)
 - Network Address: 172.16.0.128
 - Valid Host Range: 172.16.0.129 – 172.16.0.190
 - Broadcast Address: 172.16.0.191
 - Usable Hosts: $2^6 - 2 = 62$ hosts
- IT Department (/27)
 - Network Address: 172.16.0.192
 - Valid Host Range: 172.16.0.193 – 172.16.0.222
 - Broadcast Address: 172.16.0.223
 - Usable Hosts: $2^5 - 2 = 30$ hosts
- Admin Department (/28)
 - Network Address: 172.16.0.224
 - Valid Host Range: 172.16.0.225 – 172.16.0.238
 - Broadcast Address: 172.16.0.239
 - Usable Hosts: $2^4 - 2 = 14$ hosts

Remaining Unused IP Range

The allocated address space spans from 172.16.0.0 to 172.16.0.239.

- Unused IP Address Range: 172.16.0.240 to 172.16.255.255

4. A company uses the IP address 172.16.10.0/24 and has implemented subnetting using a /27 subnet mask for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address 172.16.10.78 belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address 172.16.10.95 falls within the same subnet range based on the given configuration.

ANSWER :-

Fixed-Length Subnet Masking (FLSM)

- Number of Subnets: 2^S , where S is the number of borrowed subnet bits. Here, borrowing from /24 to /27 gives $S = 27 - 24 = 3$ bits.
- Subnet Interval (Block Size): $2^{(32-27)} = 32$.

Calculations & Results

- Subnets Created: $2^3 = 8$ subnets
- Usable Hosts per Subnet: $2^5 - 2 = 30$ hosts
- Subnet Analysis for 172.16.10.78:
 - Subnet multiples of 32: 0, 32, 64, 96...
 - Since $64 \leq 78 \leq 95$, IP 172.16.10.78 belongs to the 3rd subnet block:
 - Network Address: 172.16.10.64
 - Broadcast Address: 172.16.10.95
 - Valid Host Range: 172.16.10.65 – 172.16.10.94
- Analysis for IP 172.16.10.95:
 - 172.16.10.95 falls within the numeric boundary of the 172.16.10.64/27 subnet, but it is the broadcast address. Therefore, it cannot be assigned as a valid host IP.

5.A network uses the Internet Protocol version 6 address

2001:0db8:0000:0000:0000:ff00:0042:8329. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of /64, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

ANSWER :-

IPv6 Addressing Rules & Calculations

1. Leading Zero Compression: Leading zeros within any 16-bit hexadecimal group (hextet) can be omitted (e.g., :0000: becomes :0:).
2. Double Colon Substitution (::): A single contiguous sequence of all-zero hextets can be replaced by ::. This can only be applied once per address to prevent ambiguity.

Step-by-Step Processing

- Given Address: 2001:0db8:0000:0000:0000:ff00:0042:8329

1. Shortest Compressed Form:

- Remove leading zeros: 2001:db8:0:0:0:ff00:42:8329
- Replace three consecutive :0: groups with :: 2001:db8::ff00:42:8329

2. Expanded Form:

- Restore 8 hextets by expanding :: and padding leading zeros to 4 hex digits each:

2001:0db8:0000:0000:0000:ff00:0042:8329

3. Network Prefix (/64 Prefix Length):

- The first 64 bits (first 4 hextets) form the network prefix: 2001:db8::/64 (or 2001:0db8:0000:0000::/64)

4. Total Possible Host Addresses:

- In IPv6, host bit capacity is $128 - \text{Prefix Length} = 128 - 64 = 64$ bits.
- Total possible host addresses: $2^{64} = 18,446,744,073,709,551,616$ addresses (approximately 18.44×10^{18}).